

Jing Shi

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Current Position

2022–Present **Research Scientist**, *Adobe Research*, San Jose

Research Interests

- **Post-training and self-improvement**: reinforcement learning, self-play, and feedback loops for vision-language models.
- **Evaluation and judgment**: multimodal judges, reward models, and evaluation protocols.
- **Model architecture**: learning dynamics, efficient architectures, and inference-time behavior.

Education

2017–2022 **Ph.D.**, *University of Rochester*, Rochester
Computer Science. Advisor: *prof. Chenliang Xu*.

2013–2017 **Bachelor**, *University of Electronic Science and Technology of China*, Chengdu,
GPA:3.97/4.00 (92.2/100) Ranking:1/209
Automation Engineering

Work & Research Experience

May.2021– Sep.2021	Adobe Research , Research Intern, Mentor: Ning Xu	<i>San Jose, CA, USA</i> ,
May.2019– Sep.2019	Adobe Research , Research Intern, Mentor: Ning Xu	<i>San Jose, CA, USA</i> ,
May.2018– Sep.2018	Tencent AI Lab , Research Intern, Mentor: Jia Xu, Boqing Gong	<i>Shenzhen, Guangdong, China</i> ,
Sep.2017– Sep.2022	UR Vision Group , Research Assitant, Advisor: <i>prof. Chenliang Xu</i>	<i>Rochester, NY, USA</i> ,
Mar.2016– July.2017	UESTC Rototics Center , Research Assitant, Advisor: <i>prof. Wei He</i>	<i>Chengdu, Sichuan, China</i> ,

Service

2026 Area Chair, NeurIPS
2026 Golden Reviewer, ICML

Publications

- [1] Y. Zhu, J. Shi, C. Ge, H. Tan, Y. Xu, W. Zhu, J. Kuen, K. Goswami, R. Jain, Y. Chen, M. Tao, and J. Gu, “FLARE: Diffusion for hybrid language model,” *arXiv preprint arXiv:2606.01774*, 2026. Equal contribution.

- [2] J. Yang, D. Zhang, X. Song, Q. Dai, X. Liu, Y. Chen, A. Vashishtha, J. Shi, C. Tan, and H. Peng, "CausaLab: A scalable environment for interactive causal discovery toward ai scientists," *arXiv preprint arXiv:2605.26029*, 2026.
- [3] H.-Y. Lee, H. Jiang, Y. Mei, J. Shi, M.-H. Yang, and Z. Shu, "MotiMotion: Motion-controlled video generation with visual reasoning," in *International Conference on Machine Learning (ICML)*, 2026.
- [4] Q. Wu, J. Shi, S. Jenni, K. Kafle, T. Wang, S. Chang, and H. Zhao, "RETOUCHIQ: MLLM agents for instruction-based image retouching with generalist reward," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026.
- [5] Y. Savani, B. Kveton, Y. Liu, Y. Wang, J. Shi, S. Mukherjee, N. Vlassis, and K. K. Singh, "Stepwise credit assignment for GRPO on flow-matching models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026.
- [6] T. Nguyen, S. Mo, K. K. Singh, Y. Wang, J. Shi, N. Kolkin, E. Shechtman, Y. J. Lee, and Y. Li, "Relational visual similarity," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026.
- [7] Q. Wang, B. Liu, T. Zhou, J. Shi, Y. Lin, Y. Chen, H. H. Li, K. Wan, and W. Zhao, "VISION-ZERO: Scalable VLM self-evolution via multi-agent self-play," in *International Conference on Learning Representations (ICLR)*, 2026.
- [8] J. Lu, S. Jenni, K. Kafle, J. Shi, H. Zhao, and Y. Fu, "Seeing through words: Controlling visual retrieval quality with language models," in *International Conference on Learning Representations (ICLR)*, 2026.
- [9] Z. Xiong, W. Xiong, J. Shi, H. Zhang, Y. Song, and N. Jacobs, "GroundingBooth: Grounding text-to-image customization," *Transactions on Machine Learning Research*, 2026.
- [10] J. Yuan, H. Li, X. Ding, W. Xie, Y.-J. Li, W. Zhao, K. Wan, J. Shi, X. Hu, and Z. Liu, "Understanding and mitigating numerical sources of nondeterminism in llm inference," in *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025.
- [11] A. Black, J. Shi, Y. Fan, and J. Collomosse, "ImProvShow: Multimodal fusion for image provenance summarization," in *British Machine Vision Conference (BMVC)*, 2025.
- [12] H. A. Just, Y. Fan, H. Zhao, J. Gu, R. Zhang, S. Jenni, K. Kafle, R. Jia, and J. Shi, "More than the final answer: Improving visual extraction and logical consistency in vision-language models," *arXiv preprint arXiv:2512.12487*, 2025.
- [13] K. Akdemir, J. Shi, K. Kafle, B. Price, and P. Yanardag, "Plot'n Polish: Zero-shot story visualization and disentangled editing with text-to-image diffusion models," *arXiv preprint arXiv:2509.04446*, 2025.

- [14] J. Hernandez, J. Shi, S. Jenni, V. Ordonez, and K. Kafle, "Improving large vision and language models by learning from a panel of peers," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 1402–1412, 2025.
- [15] Z. Di, J. Shi, Y. Fan, H. Tan, A. Black, J. Collomosse, and Y. Liu, "DiffTell: A high-quality dataset for describing image manipulation changes," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 24580–24590, 2025.
- [16] K. Wang, S. Deng, J. Shi, D. Hatzinakos, and Y. Tian, "Av-dit: Taming image diffusion transformers for efficient joint audio and video generation," in *Proceedings of the 33rd ACM International Conference on Multimedia*, pp. 10486–10495, 2025.
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- [18] J. Nam, S. Son, Z. Xu, J. Shi, D. Liu, F. Liu, S. Kim, and Y. Zhou, "Visual persona: Foundation model for full-body human customization," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 18630–18641, 2025.
- [19] D. Qi, H. Zhao, J. Shi, S. Jenni, Y. Fan, F. Derroncourt, S. Cohen, and S. Li, "The photographer's eye: Teaching multimodal large language models to see, and critique like photographers," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 24807–24816, 2025.
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- [23] H. Hua, J. Shi, K. Kafle, S. Jenni, D. Zhang, J. Collomosse, S. Cohen, and J. Luo, "Finematch: Aspect-based fine-grained image and text mismatch detection and correction," in *European Conference on Computer Vision*, pp. 474–491, Springer, 2024.
- [24] Y. Ren, Y. Zhou, J. Yang, J. Shi, D. Liu, F. Liu, M. Kwon, and A. Shrivastava, "Customize-a-video: One-shot motion customization of text-to-video diffusion models," in *European Conference on Computer Vision*, pp. 332–349, Springer, 2024.

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